

AOS-CX 10.06 SNMP/MIB Guide

6100, 6200, 6300, 6400, 8320, 8325, 8360, 8400 Switch Series



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Applicable products

This document applies to the following products:

- Aruba 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)
- Aruba 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A)
- Aruba 6300 Switch Series (JL658A, JL659A, JL660A, JL661A, JL662A, JL663A, JL664A, JL665A, JL666A, JL667A, JL668A, JL762A)
- Aruba 6400 Switch Series (JL741A, R0X26A, R0X27A, R0X29A, R0X30A)
- Aruba 8320 Switch Series (JL479A, JL579A, JL581A)
- Aruba 8325 Switch Series (JL624A, JL625A, JL626A, JL627A)
- Aruba 8360 Switch Series (JL700A, JL701A, JL702A, JL703A, JL706A, JL707A, JL708A, JL709A, JL710A, JL711A)
- Aruba 8400 Switch Series (JL375A, JL376A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and other resources](#).

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the interface context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where *<VLAN-ID>* is a variable representing the VLAN number.

Switch prompts in examples

The switch prompts used in this document are examples and might not match your particular switch or environment.

In examples:

- The switch prompt starts with the word `switch`.
- The switch prompt also indicates the command context.

For example:

switch>

Indicates the operator command context.

switch#

Indicates the manager command context.

switch(config)#

Indicates the global configuration context.

In your environment, the switch prompt can vary because the prompt is user-configurable.

- Typically, the switch prompt begins with the host name of the switch.
- The switch prompt contains specifier in certain configuration command contexts, such as interface name or VLAN ID. For example: `switch(config-vlan-100) #`

In these cases, examples in this document might contain placeholders such as `n` or `if`.

Simple Network Management Protocol (SNMP) is an Internet-standard protocol used for managing and monitoring the devices connected to a network by collecting, organizing and modifying information about managed devices on IP networks.

Configuring SNMP

The SNMP agent provides read-only access in this release.

Procedure

1. SNMP is not enabled on the switch by default, unless the user enables it over any available VRF or with the default/mgmt VRF using the command `snmp-server vrf <NAME>`. For example, use the command `snmp-server vrf mgmt` to enable SNMP on the management interface. Use the command `snmp-server vrf default` to enable SNMP on the default VRF. Use the command `snmp-server vrf <USERDEFINED_VRF_NAME>` to enable SNMP on the user created VRF.
2. Set the system contact, location, and description for the switch with the following commands:
 - `snmp-server system-contact`
 - `snmp-server system-location`
 - `snmp-server system-description`
3. If required, change the default SNMP port on which the agent listens for requests with the command `snmp-server agent-port`.
4. By default, the agent uses the community string **public** to protect access through SNMPv1/v2c. Set a new community string with the command `snmp-server community`.
5. Configure the trap receivers to which the SNMP agent will send trap notifications with the command `snmp-server host`.
6. Create an SNMPv3 context and associate it with any available SNMPv3 user to perform context specific v3 MIB polling using the command `snmpv3 user <V3_USERNAME> context <CONTEXT_NAME>`.
7. Create an SNMPv3 context and associate it with an available SNMPv1/v2c community string to perform context specific v1/v2c MIB polling using the command `snmpv3 context <CONTEXT_NAME> vrf <VRF_NAME> community <COMMUNITY_NAME>`.
8. Review your SNMP configuration settings with the following commands:
 - `show snmp agent-port`
 - `show snmp community`
 - `show snmp system`
 - `show snmpv3 context`

- `show snmp trap`
- `show snmp vrf`
- `show snmpv3 users`
- `show tech snmp`

Example

Example 1

This example creates the following configuration:

- Enables SNMP on the out-of-band management interface (VRF `mgmt`).
- Sets the contact, location, and description for the switch to: **JaniceM, Building2, LabSwitch**.
- Sets the community string to **Lab8899X**.

```
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server system-contact JaniceM
switch(config)# snmp-server system-location Building2
switch(config)# snmp-server system-description LabSwitch
switch(config)# snmp-server community Lab8899X
```

Example 2

This example creates the following configuration:

- Creates an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**.
- Associates the SNMPv3 user `Admin` with a context named `newContext`.

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
priv-pass plaintext myprivpass
switch(config)# snmpv3 user Admin context newContext
```

SNMP commands

event-trap-enable

Syntax

```
event-trap-enable
```

```
no event-trap-enable
```

Description

Enables the notification of events to be sent as traps to the SNMP management stations. It is enabled by default.

The `no` form of this command disables the event traps.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the event traps:

```
switch(config)# event-trap-enable
```

Disabling the event traps:

```
switch(config)# no event-trap-enable
```

rmon alarm

Syntax

```
rmon alarm index <INDEX> snmp-oid <SNMP-OID> rising-threshold <RISING-THRESHOLD>  
    falling-threshold <FALLING-THRESHOLD> [sample-interval <SAMPLE-INTERVAL>] [sample-type <ABSOLUTE|DELTA>]  
  
no rmon alarm [index <INDEX>]
```

Description

Stores configuration entries in an alarm table that defines the sample interval, sample-type, and threshold parameters for an SNMP MIB object. Only the SNMP MIB objects that resolve to an ASN.1 primitive type of INTEGER (INTEGER, Integer32, Counter32, Counter64, Gauge32, or TimeTicks) will be monitored.

The **no** form of this command removes all RMON alarms and allows you to specify an index to remove a particular RMON alarm.

Command context

config

Parameters

index <INDEX>

Specifies the RMON alarm index. Range: 1 to 20.

snmp-oid <SNMP-OID>

Specifies the SNMP MIB object to be monitored by RMON.

rising-threshold <RISING-THRESHOLD>

Specifies the upper threshold value for the RMON alarm.

falling-threshold <FALLING-THRESHOLD>

Specifies the falling threshold value for the RMON alarm. The falling threshold must be less than the rising threshold.

sample-interval <SAMPLE-INTERVAL>

Sample interval in seconds. Default: 30.

sample-type <ABSOLUTE|DELTA>

Specifies the method of sampling of the SNMP MIB object. Default: Absolute.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring RMON for the MIB object `ifOutErrors.15` with an index `1`, rising threshold of `2147483647` and falling threshold of `-2134` using **absolute** sampling for a sample interval of `100` seconds:

```
switch(config)# rmon alarm index 1 snmp-oid ifOutErrors.15 rising-threshold 2147483647
falling-threshold -2134 sample-type absolute sample-interval 100
```

Removing RMON alarm with the index `5`:

```
switch(config)# no rmon alarm index 5
```

```
rmon alarm {enable | disable} {index | all}
```

Syntax

```
rmon alarm {enable | disable} {index <INDEX> | all}
```

Description

Enables and disables the RMON alarm and its index. RMON alarm is enabled by default.

Command context

config

Parameters

enable

Enables the RMON alarm index

disable

Disables the RMON alarm index.

index <INDEX>

Specifies the RMON alarm index. Range: 1 to 20.

all

Specifies all the RMON alarms.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling or disabling all the RMON alarm:

```
switch(config)# rmon alarm enable all
switch(config)# rmon alarm disable all
```

Enabling or disabling RMON alarm by index:

```
switch(config)# rmon alarm enable index 1
switch(config)# rmon alarm disable index 1
```

show rmon alarm

Syntax

```
show rmon alarm [index <INDEX>]
```

Description

Displays the RMON alarm configurations.

Command context

config

Parameters

index <INDEX>

Specifies the RMON alarm index. Range: 1 to 20.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing all RMON alarm configurations:

```
switch# show rmon alarm
Index          : 1
Enabled        : true
Status         : valid
MIB object     : ifOutErrors.15
Sample type    : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-09-21 05:58:11

Index          : 3
Enabled        : true
Status         : invalid
MIB object     : IF-MIB::ifDescr.19
Sample type    : absolute
Sampling interval : 10000 seconds
Rising threshold : 4000
Falling threshold : 10
Last sampled value : 0
```

Showing RMON alarm with alarm index 1:

```
switch# show rmon alarm index 1
Index          : 1
Enabled        : true
Status         : valid
MIB object     : ifOutErrors.15
Sample type    : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-06-21 05:58:11
```

Showing disabled RMON alarm information:

```
switch# show rmon alarm
Index          : 1
Enabled        : false
Status         : valid
MIB object     : ifOutErrors.15
Sample type    : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-09-21 05:58:11

Index          : 3
Enabled        : false
Status         : invalid
MIB object     : IF-MIB::ifDescr.19
Sample type    : absolute
Sampling interval : 10000 seconds
Rising threshold : 4000
Falling threshold : 10
Last sampled value : 0
```

show snmp agent-port

Syntax

```
show snmp agent-port
```

Description

Displays SNMP agent UDP port number.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying SNMP agent UDP port number:

```
switch# show snmp agent-port
SNMP agent port : 161
```

show snmp community

Syntax

```
show snmp community
```

Description

Displays a list of all configured SNMPv1/v2c communities.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

When a user creates a custom community before enabling an SNMP agent, AOS-CX automatically removes the default `public` community from the system.

Example

Displaying a list of all configured SNMPv1/v2c communities:

Before any community is created by user

```
switch# show snmp community
-----
SNMP communities
-----
public
```

After community is created by user

```
switch#show snmp community
-----
SNMP communities
-----
private
private2
```

show snmp system

Syntax

```
show snmp system
```

Description

Displays SNMP description, location, and contact information.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying SNMP description, location, and contact information:

```
switch# show snmp system
SNMP system information
-----
System description : Aggregation router
System location   : Main lab
System contact    : John Smith, Lab Admin
```

show snmp trap

Syntax

```
show snmp trap
```

Description

Displays all configured SNMP traps/informs receivers.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying all configured SNMP trap and informs receivers:

```
switch# show snmp trap
HOST                                PORT  TYPE  VER  COMMUNITY/USER NAME  VRF
-----
10.10.10.10                        162   trap  v1   public                default
10.10.10.10                        162   inform v2c public                default
10.10.10.10                        162   inform v3 name          default
```

show snmp vrf

Syntax

```
show snmp vrf
```

Description

Displays the VRF on which the SNMP agent service is running.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying SNMP services enabled on VRF:

```
switch# show snmp vrf
SNMP enabled VRF
-----
mgmt
default
```

show snmpv3 context

Syntax

```
show snmpv3 context
```

Description

Displays all configured SNMP contexts.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying all configured SNMP contexts:

```
switch# show snmpv3 context
```

name	vrf	community
contextA	default	private
contextB	vrf_A	public

```
switch# show snmpv3 context
```

Name	vrf	Community	ype[Instance_id]
A	default	public	vrf

```
switch#
```

show snmpv3 engine-id

Syntax

```
show snmpv3 engine-id
```

Description

Displays the configured SNMPv3 snmp engine-id.

If the SNMPv3 engine-id is not configured, by default a unique engine-id is created by the switch using a combination of the enterprise OID value and the switch's mac address.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying the configured SNMPv3 engine-id:

```
switch# show snmpv3 engine-id
SNMP engine-id : 80:00:B8:5C:08:00:09:1d:de:a5
```

show snmpv3 security-level

Syntax

```
show snmpv3 security-level
```

Description

Displays the configured SNMPv3 security level.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying the configured SNMPv3 security level:

```
switch# show snmpv3 security-level
SNMPv3 security-level : auth
```

show snmpv3 users

Syntax

```
show snmpv3 users
```

Description

Displays all configured SNMPv3 users.

For more details on the user enabled status, see [snmpv3 security-level](#).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying all configured SNMPv3 users:

```
switch# show snmpv3 users
```

User	AuthMode	PrivMode	Context	Enabled
name	md5	none	none	False
name2	sha	aes	none	True

snmp-server agent-port

Syntax

```
snmp-server agent-port <PORT>
```

```
no snmp-server agent-port [<PORT>]
```

Description

Sets the UDP port number that the SNMP master agent uses to communicate. UDP port 161 is the default port.

The `no` form of this command sets the SNMP master agent port to the default value.

Command context

```
config
```

Parameters

<PORT>

Specifies the UDP port number that the SNMP master agent will use. Range: 1 to 65535. Default: 161.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the SNMP master agent port to 2000:

```
switch(config)# snmp-server agent-port 2000
```

Resetting the SNMP master agent port to the default value:

```
switch(config-schedule)# no snmp-server agent-port 2000
```

snmp-server community

Syntax

```
snmp-server community <STRING>
```

```
no snmp-server community <STRING>
```

Description

Adds an SNMPv1/SNMPv2c community string. A community string is a password that controls read access to the SNMP agent. A network management program must supply this name when attempting to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The `no` form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value `public` is automatically defined.

Command context

```
config
```


Parameters

<STRING>

Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config)# snmp-server community private
```

Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config)# no snmp-server community private
```

snmp-server historical-counters-monitor

Syntax

```
snmp-server historical-counters-monitor
```

```
no snmp-server historical-counters-monitor
```

Description

Enables the Remote Network Monitoring agent (`rmond`) to start collecting historical interface statistics. The `no` form of this command stops the historical interface statistics collection.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Enabling the `rmond` agent to start historical interface statistics collection:

```
switch(config)# snmp-server historical-counters-monitor
```

Disabling the `rmond` agent to stop historical interface statistics collection:

```
switch(config)# no snmp-server historical-counters-monitor
```

snmp-server host

Syntax

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community <STRING>]  
[port <UDP-PORT>] [<VRF-NAME>]
```

```
no snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community <STRING>]  
[port <UDP-PORT>] [<VRF-NAME>]
```

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>]

no snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>]

snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [<VRF-NAME>]

no snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [<VRF-NAME>]
```

Description

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The **no** form of this command removes the specified trap/inform receiver.

Command context

config

Parameters

<IPv4-ADDR>

Specifies the IP address of a trap receiver in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100.

<IPv6-ADDR>

Specifies the IP address of a trap receiver in IPv6 format (x:x::x:x).

trap version <VERSION>

Specifies the trap notification type for SNMPv1, v2c or v3. Available options are: v1, v2c or v3.

inform version v2c

Specifies the inform notification type for SNMPv2c.

trap version v3

Specifies the trap notification type for SNMPv3.

user <NAME>

Specifies the SNMPv3 user name to be used in the SNMP trap notifications.

community <STRING>

Specifies the name of the community string to use when sending trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: public.

<UDP-PORT>

Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.

<VRF-NAME>

Specifies the VRF on which the SNMP agent listens for incoming requests.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# snmp-server host 10.10.10.10 trap version v1
switch(config)# no snmp-server host 10.10.10.10 trap version v1

switch(config)# snmp-server host a:b::c:d trap version v1
switch(config)# no snmp-server host a:b::c:d trap version v1

switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public

switch(config)# snmp-server host a:b::c:d trap version v2c community public
switch(config)# no snmp-server host a:b::c:d trap version v2c community public

switch(config)# snmp-server host 10.10.10.10 trap version v2c community public port 5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public port 5000

switch(config)# snmp-server host 10.10.10.10 trap version v2c community public port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public port 5000 vrf default

switch(config)# snmp-server host a:b::c:d trap version v2c community public port 5000
switch(config)# no snmp-server host a:b::c:d trap version v2c community public port 5000

switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public

switch(config)# snmp-server host a:b::c:d inform version v2c community public
switch(config)# no snmp-server host a:b::c:d inform version v2c community public

switch(config)# snmp-server host 10.10.10.10 inform version v2c community public port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public port 5000

switch(config)# snmp-server host 10.10.10.10 inform version v2c community public port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public port 5000 vrf default

switch(config)# snmp-server host a:b::c:d inform version v2c community public port 5000
switch(config)# no snmp-server host a:b::c:d inform version v2c community public port 5000

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin port 2000

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin port 2000
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin port 2000
```

snmp-server system-contact

Syntax

```
snmp-server system-contact <INFO>
```

```
no snmp-server system-contact [<INFO>]
```

Description

Sets SNMP contact information.

The `no` form of this command removes the SNMP contact information.

Command context

```
config
```

Parameters

<INFO>

Specifies SNMP contact information. Range: 1 to 128 printable ASCII characters, except for question mark (?).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defines SNMP contact information to be **John Smith, Lab Admin**:

```
switch(config)# snmp-server system-contact John Smith, Lab Admin
```

Removes SNMP contact information:

```
switch(config)# no snmp-server system-contact
```

snmp-server system-description

Syntax

```
snmp-server system-description <DESCRIPTION>
```

```
no snmp-server system-description
```

Description

Sets the SNMP system description.

The **no** form of this command removes the SNMP system description.

Command context

```
config
```

Parameters

<DESCRIPTION>

Specifies the SNMP system description. Typical content to include would be the full name and version of the following:

- Hardware type of the system
- Software operating system
- Networking software

Range: 1 to 64 printable ASCII characters, except for the question mark (?).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defines the SNMP system description to be **mainSwitch**:

```
switch(config)# snmp-server system-description mainSwitch
```

Removes the SNMP system description:

```
switch(config)# no snmp-server system-description mainSwitch
```

snmp-server system-location

Syntax

```
snmp-server system-location <INFO>
```

```
no snmp-server system-location
```

Description

Sets the SNMP location information.

The `no` form of this command removes the SNMP location information.

Command context

config

Parameters

<INFO>

Specifies the SNMP location information. Range: 1 to 128 printable ASCII characters, except for the question mark (?).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defines the SNMP location information to be **Main Lab**:

```
switch(config)# snmp-server system-location Main Lab
```

Removes the SNMP location information:

```
switch(config)# no snmp-server system-location
```

snmp-server vrf

Syntax

```
snmp-server vrf <VRF-NAME>
```

```
no snmp-server vrf <VRF-NAME>
```

Description

Configures a VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF. 6100 only supports default VRF. The SNMP agent can listen on multiple VRFs.

The `no` form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

Command context

config

Parameters

<VRF-NAME>

Specifies the name of a VRF.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the SNMP agent to listen on VRF default.

```
switch(config)# snmp-server vrf default
```

Configuring the SNMP agent to listen on VRF mgmt.

```
switch(config)# snmp-server vrf mgmt
```

Configuring the SNMP agent to listen on user-defined VRF myvrf.

```
switch(config)# snmp-server vrf myvrf
```

Stopping the SNMP agent from listening on VRF default.

```
switch(config)# no snmp-server vrf default
```

snmp-server trap-source interface vrf

Syntax

```
snmp-server trap-source {interface <IF-NAME> | <IPv4-Address> | <IPv6-Address>} [vrf <VRF-NAME>]
```

```
no snmp-server trap-source {interface <IF-NAME> | <IPv4-Address> | <IPv6-Address>} [vrf <VRF-NAME>]
```

Description

Configures SNMP trap source interface or IP address for a VRF.

The **no** form of this command removes the SNMP **trap-source** configuration for a VRF.

Command context

```
config
```

Parameters

<IF-NAME>

Specifies the source interface name. Interface name can be physical interface, loopback interface, LAG interface, or VLAN interface.

<IPv4-Address>

Specifies the IPv4 address of source interface for the SNMP trap.

<IPv6-Address>

Specifies the IPv6 address of source interface for the SNMP trap.

<VRF-NAME>

Specifies the name of a VRF associated to the source interface for the SNMP trap.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring SNMP trap source interface for a VRF.

```
switch(config)# snmp-server trap-source interface 1/1/12 vrf sample
switch(config)# snmp-server trap-source interface loopback10 vrf sample
switch(config)# snmp-server trap-source interface vlan23 vrf sample
```

Configuring SNMP trap source IP address for a VRF.

```
switch(config)# snmp-server trap-source 10.0.0.1 vrf red
switch(config)# snmp-server trap-source 1001::1 vrf red
```

snmpv3 context

Syntax

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]
```

```
no snmpv3 context <NAME> [vrf <VRF-NAME>]
```

Description

Creates an SNMPv3 context on the specified VRF.

The `no` form of this command removes the specified SNMP context.

Command context

config

Parameters

<NAME>

Specifies the name of the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark (?).

vrf <VRF-NAME>

Specifies the VRF associated with the context. Default: default.

community <STRING>

Specifies the SNMP community string associated with the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating an SNMPv3 context named `newContext`:

```
switch(config)# snmpv3 context newContext
```

Creating an SNMPv3 context named `newContext` on VRF `myVrf` and with community string `private`.

```
switch(config)# snmpv3 context newContext vrf myVrf community private
```

Removing the SNMPv3 context named **newContext** on VRF **myVrf**:

```
switch(config)# no snmpv3 context newContext vrf myVrf
```

snmpv3 engine-id

Syntax

```
snmpv3 engine-id <ENGINE-ID>
```

```
no snmpv3 engine-id <ENGINE-ID>
```

Description

Configures the SNMPv3 SNMP engine-id allowing an administrator to configure a unique SNMP engine-id for the switch. This engine-id is used by the NMS management tool to identify and distinguish multiple switches on the same network.

The **no** form of this command restores the default engine-id, created by the switch using a combination of the enterprise OID value and the switch's mac address.

Command context

```
config
```

Parameters

<ENGINE-ID>

SNMPv3 SNMP engine-id in colon separated hexadecimal notation.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the SNMPv3 engine-id:

```
switch(config)#  
switch(config)# snmpv3 engine-id  
WORD SNMPv3 snmp engine-id in colon seperated hexadecimal notation  
switch(config)# snmpv3 engine-id 01:23:45:67:89:ab:cd:ef:01:23:45:67
```

Restoring the default SNMPv3 engine-id:

```
switch(config)# no snmpv3 engine-id
```

snmpv3 security-level

Syntax

```
snmpv3 security-level {auth | auth-privacy}
```

```
no snmpv3 security-level {auth | auth-privacy}
```

Description

Configures the SNMPv3 security level. The security level determines which SMNPv3 users defined by the command **snmpv3 user** are able to connect.

The **no** form of this command changes the security level as follows:

- `no snmpv3 security-level auth`: Sets the security level to auth-privacy.
- `no snmpv3 security-level auth-privacy`: Sets the security level to no authentication or privacy, allowing any SNMP user to connect.

Command context

config

Parameters

auth

SNMPv3 users that support authentication, or authentication and privacy are allowed.

auth-privacy

Only SNMPv3 users with both authentication and privacy are allowed. This is the highest level of SNMPv3 security. Default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the SNMPv3 security level to authentication and privacy:

```
switch(config)# snmpv3 security-level auth-privacy
```

Setting the SNMPv3 security level to authentication only:

```
switch(config)# snmpv3 security-level auth
```

Setting the SNMPv3 security level to no authentication and no privacy:

```
switch(config)# no snmpv3 security-level auth-privacy
```

Restoring the default SNMPv3 security level to authentication and privacy:

```
switch(config)# no snmpv3 security-level auth
```

snmpv3 user

Syntax

```
snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass {plaintext | ciphertext}
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass {plaintext | ciphertext} <PRIV-PWORD>] ]
```

```
no snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass <PRIV-PWORD>] ]
```

Description

Creates an SNMPv3 user and adds it to an SNMPv3 context. The `no` form of this command removes the specified SNMPv3 user.

For more details on the user enabled status, see [snmpv3 security-level](#).

Command context

config

Parameters

<NAME>

Specifies the SNMPv3 username. Range 1 to 32 printable ASCII characters, excluding space and question mark (?).

auth <AUTH-PROTOCOL>

Specifies the authentication protocol used to validate user logins. Available options are: md5 or sha.

auth-pass {plaintext | ciphertext} <AUTH-PWORD>

Specifies the SNMPv3 user password. Range for plaintext is 8 to 32 printable ASCII characters, excluding space and question mark (?).

Range for ciphertext is 1 to 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.

priv <PRIV-PROTOCOL>

Specifies the SNMPv3 security protocol (encryption method). Available options are: aes or des.

priv-pass {plaintext | ciphertext} <PRIV-PWORD>

Specifies the SNMPv3 user privacy passphrase. Range for plaintext is 8 to 32 printable ASCII characters, excluding space and question mark (?).

Range for ciphertext is 1 to 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des priv-pass plaintext myprivpass
```

Removing an SNMPv3 user named **Admin**:

```
switch(config)# no snmpv3 user Admin
```

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des priv-pass plaintext myprivpass
```

Copying an SNMP user from switch 1 to switch 2.

On switch 1, configure a user called **Admin**, then issue the `show running-config` command to display switch configuration settings. The `snmpv3 user` command uses the `ciphertext` option to protect the users's passwords.

```
switch1(config)# snmpv3 user Admin auth sha auth-pass ciphertext mypassword
priv des priv-pass ciphertext myprivpass
switch1(config)# exit
switch1# show running-config
Current configuration:
!
!Version ArubaOS-CX XL.10.04.0001AD
!
!
```

```

snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
ssh server vrf mgmtdefault
!
!
!
!
interface mgmtinterface vlan 1
    no shutdown
    ip dhcp
vlan 1

```

On switch 2, execute the `snmpv3 user` command that was displayed by `show running-config` on switch 1. This creates the user on switch 2 with the same configuration settings.

```

switch1(config)# snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=

```

The Entity MIB, rfc 6933, allows network managers to retrieve physical containment and logical relationships for devices in the network. The entconfigChange trap is sent to configured SNMP-server hosts when a change occurs. The trap is configured to send notifications no more than once every 5 seconds. We will be supporting the Entity MIB for read-only.

Physical components that are supported include:

- Chassis
- Fabric cards
- Fan trays
- Fans
- Line cards and their interfaces
- Management modules and the intake temperature sensor
- Power supplies

The slots for any removable component are also represented.

The logical table of the Entity MIB represents configured VLANs and the associated ports.

entConfigChange trap/notification is sent to configured snmp-server hosts.

Location of the MIB files on the web

The MIB files for Aruba switches can be found on the [Aruba Service Portal](#). You can apply the various filters to filter by product series, software versions, and software release types.

Newly introduced MIBs and Traps for 10.06

The following table contains the newly introduced MIBs and Traps for each software feature. Software is provided along with the MIB and Traps supported name.

Software feature	MIB files	Implemented MIB objects	Traps supported (Y/N)	Traps list
Networking OID	ARUBAWIRED-NETWORKING-OID	arubaWiredSwitchJL675A arubaWiredSwitchJL676A arubaWiredSwitchJL677A arubaWiredSwitchJL678A arubaWiredSwitchJL679A arubaWiredSwitchModuleJL675A arubaWiredSwitchModuleJL676A arubaWiredSwitchModuleJL677A arubaWiredSwitchModuleJL678A arubaWiredSwitchModuleJL679A arubaWiredSwitchJL717A arubaWiredSwitchJL718A arubaWiredSwitchJL719A arubaWiredSwitchJL720A arubaWiredSwitchJL721A arubaWiredSwitchJL722A arubaWiredSwitchModuleJL717A arubaWiredSwitchModuleJL718A arubaWiredSwitchModuleJL719A arubaWiredSwitchModuleJL720A arubaWiredSwitchModuleJL721A arubaWiredSwitchModuleJL722A arubaWiredSwitchNonPoEPowerSupplyUnit arubaWiredSwitchPoEPowerSupplyUnit arubaWiredSwitch8360FanTraySlot arubaWiredSwitch8360PowerSupplySlot arubaWiredSwitchPowerSupplyUnitJL600A	N	NA

Table Continued

Software feature	MIB files	Implemented MIB objects	Traps supported (Y/N)	Traps list
		arubaWiredSwitchPowerSupplyUnitJL601A arubaWiredSwitchPowerSupplyUnitJL712A arubaWiredSwitchPowerSupplyUnitJL713A arubaWiredSwitchFanTrayJL714A arubaWiredSwitchFanTrayJL715A arubaWiredSwitch36WPowerSupplyUnit arubaWiredSwitch65WPowerSupplyUnit arubaWiredSwitch165WPowerSupplyUnit arubaWiredSwitch500WPowerSupplyUnit		
MDNS	ARUBAWIRED-MDNS-MIB	arubaWiredMdnsAdminState arubaWiredMdnsServicetable arubaWiredMdnsServiceName arubaWiredMdnsServiceIdIndex arubaWiredMdnsServiceId arubaWiredMdnsProfileTable arubaWiredMdnsProfileName arubaWiredMdnsProfileVIDList arubaWiredMdnsProfilePermitCount arubaWiredMdnsProfileDenyCount arubaWiredMdnsProfileFilterRuleTable arubaWiredMdnsProfileFilterRuleProfileName arubaWiredMdnsProfileFilterRuleIndex arubaWiredMdnsProfileFilterRuleServiceName arubaWiredMdnsProfileFilterRuleInstanceName	N	NA

Table Continued

Software feature	MIB files	Implemented MIB objects	Traps supported (Y/N)	Traps list
		arubaWiredMdnsProfileFilterRuleAction arubaWiredMdnsPortTable arubaWiredMdnsPortVlanId arubaWiredMdnsPortMdnsEnabled arubaWiredMdnsPortTxProfileName arubaWiredMdnsPortRxProfileName arubaWiredMdnsServiceProviderTable arubaWiredMdnsServiceProviderServiceId arubaWiredMdnsServiceProviderServiceIdIndex arubaWiredMdnsServiceProviderServiceInstanceName arubaWiredMdnsServiceProviderVlanId arubaWiredMdnsServiceProviderMacAddress arubaWiredMdnsServiceProviderHostName arubaWiredMdnsServiceProviderIpAddress arubaWiredMdnsServiceProviderTtl arubaWiredMdnsServiceProviderExpireTime		

Table Continued

Software feature	MIB files	Implemented MIB objects	Traps supported (Y/N)	Traps list
MSTP	ARUBAWIRED-MSTP-MIB	arubaWiredMstpBpduGuardTimeout arubaWiredMstpPortTable arubaWiredMstpPortIndex arubaWiredMstpPortAdminEdge arubaWiredMstpPortAdminPointToPoint arubaWiredMstpPortAutoEdge arubaWiredMstpPortBpduFiltering arubaWiredMstpPortRestrictedTcn arubaWiredMstpPortRootGuard arubaWiredMstpPortLoopGuard arubaWiredMstpPortBpduProtection arubaWiredMstpPortRpvstProtection arubaWiredMstpPortRpvstFiltering	N	NA
RPVST	ARUBAWIRED-RPVST-MIB	arubaWiredRpvstBpduGuardTimeout	N	NA

Table Continued

Software feature	MIB files	Implemented MIB objects	Traps supported (Y/N)	Traps list
VSFv2	ARUBAWIRED-VSFv2-MIB	arubaWiredVsfv2TrapEnable arubaWiredVsfv2SplitDetectConfigured arubaWiredVsfv2OperStatus arubaWiredVsfv2Topology arubaWiredVsfv2StackMacAddr arubaWiredVsfv2DomainId arubaWiredVsfv2MemberTable arubaWiredVsfv2MemberIndex arubaWiredVsfv2MemberRole arubaWiredVsfv2MemberStatus arubaWiredVsfv2MemberPartNumber arubaWiredVsfv2MemberMacAddr arubaWiredVsfv2MemberProductName arubaWiredVsfv2MemberSerialNum arubaWiredVsfv2MemberBootImage arubaWiredVsfv2MemberCpuUtil arubaWiredVsfv2MemberMemoryUtil arubaWiredVsfv2MemberBootTime arubaWiredVsfv2MemberBootRomVersion arubaWiredVsfv2MemberTotalMemory arubaWiredVsfv2MemberCurrentUsage arubaWiredVsfv2LinkTable arubaWiredVsfv2LinkOperStatus arubaWiredVsfv2LinkPeerMemberId arubaWiredVsfv2LinkPeerLinkId arubaWiredVsfv2LinkPortList	Y	arubaWiredVsfv2MemberStatusChange arubaWiredVsfv2FragmentStatusChange

Table Continued

Software feature	MIB files	Implemented MIB objects	Traps supported (Y/N)	Traps list
802.1X	IEEE8021-SECY-MIB.mib	ieee8021XPaePortTable secyIfInterfaceIndex secyIfMaxPeerSCs secyIfRxMaxKeys secyIfRxMaxKeys secyIfProtectFramesEnable secyIfValidateFrames secyIfReplayProtectEnable secyIfReplayProtectWindow secyIfCurrentCipherSuite secyIfAdminPt2PtMAC secyIfOperPt2PtMAC secyIfIncludeSCIEnable secyIfUseESEnable secyIfUseSCBEnable secyIfIncludingSCI secyIfMaxTSCs secyTxSCTable secyTxSCI secyTxSCState secyTxSCEncodingSA secyTxSCEncipheringSA secyTxSCCreatedTime secyTxSCStartedTime secyTxSCStoppedTime secyTxSatable secyTxSAState secyTxSANextPN secyTxSAConfidentiality	N	NA

Table Continued

Software feature	MIB files	Implemented MIB objects	Traps supported (Y/N)	Traps list
		secyTxSASAKUnchanged secyTxSACreatedTime secyTxSASStartedTime secyTxSASStoppedTime secyRxSCTable secyRxSCState secyRxSCCurrentSA secyRxSCCreatedTime secyRxSCStartedTime secyRxSCStoppedTime secyRxSASTable secyRxSASState secyRxSANextPN secyRxSASAKUnchanged secyRxSACreatedTime secyRxSASStartedTime secyRxSASStoppedTime secyRxSANextXPN secyRxSALowestXPN secyRxSAKeyIdentifier secyRxSASSCI secyTxSASStatsTable secyTxSASStatsProtectedPkts secyTxSASStatsEncryptedPkts secyRxSASStatsTable secyRxSASStatsUnusedSAPkts secyRxSASStatsNoUsingSAPkts secyRxSASStatsNotValidPkts secyRxSASStatsInvalidPkts		

Table Continued

Software feature	MIB files	Implemented MIB objects	Traps supported (Y/N)	Traps list
		secyRxSASStatsOKPkts secyTxSCStatsTable secyTxSCStatsProtectedPkts secyTxSCStatsEncryptedPkts secyTxSCStatsOctetsProtected secyTxSCStatsOctetsEncrypted secyRxSCStatsTable secyRxSCStatsUnusedSAPkts secyRxSCStatsNoUsingSAPkts secyRxSCStatsLatePkts secyRxSCStatsNotValidPkts secyRxSCStatsInvalidPkts secyRxSCStatsDelayedPkts secyRxSCStatsUncheckedPkts secyRxSCStatsOKPkts secyStatsTable secyStatsTxUntaggedPkts secyStatsTxTooLongPkts secyStatsRxUntaggedPkts secyStatsRxNoTagPkts secyStatsRxBadTagPkts secyStatsRxUnknownSCIPkts secyStatsRxNoSCIPkts secyStatsRxOverrunPkts		
Q-BRIDGE	Q-Bridge.mib	dot1qFdbTable dot1qFdbId dot1qFdbDynamicCount dot1qTpFdbTable dot1qTpFdbAddress	N	NA

Software feature	MIB files	Implemented MIB objects	Traps supported (Y/N)	Traps list
		dot1qTpFdbPort dot1qTpFdbStatus		

Accessing Aruba Support

Aruba Support Services	https://www.arubanetworks.com/support-services/
Aruba Support Portal	https://asp.arubanetworks.com/
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
Software licensing	https://lms.arubanetworks.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/
Aruba software and documentation	https://asp.arubanetworks.com/downloads

Accessing updates

To download product updates:

Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>



IMPORTANT: Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://asp.arubanetworks.com/notifications/subscriptions> (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

Warranty information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

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